MATHEMATICS AA HL

Internal Assessment

Investigating an integral from the MIT integration bee competition

20 pages

Introduction

As someone deeply immersed in the world of mathematics and computer science competitions, mathematics has always been one of the pillars of my academic journey. This world constantly challenges me to push the boundaries of my knowledge and refine my problem-solving skills. Driven by this relentless pursuit for knowledge, I came across an interesting integral from the MIT Integration Bee (figure 1), an encounter that motivated this exploration. This prestigious competition is designed for undergraduate students, where a set number of integrals (usually five) are presented, and candidates must solve each one within four minutes. This competition often requires the use of mathematical techniques that go beyond the scope of the IB Math AA curriculum. However, it was found that the second problem presented during the MIT Integration Bee Finals 2023 could be solved using only the mathematics included in our syllabus, along with a great deal of critical thinking. More particularly, the integral can be solved using trigonometric identities, partial fractions, synthetic division, complex numbers, and rigorous calculus techniques. Therefore, the aim of this exploration is to analyse and attempt to solve this integral using advanced techniques from differential and integral calculus.

Finals Problem 2

$$\int_0^{\pi} \left(\frac{\sin(2x)\sin(3x)\sin(5x)\sin(30x)}{\sin(x)\sin(6x)\sin(10x)\sin(15x)}\right)^2 dx$$

Figure 1 - MIT Integration Bee Finals 2023

Before diving into this integral directly, my teacher suggested that we first attempt to solve three similar integrals. The approach involves starting with two terms, then progressing to four terms, six terms, and finally solving all eight terms together. This approach could help us identify repetitive patterns or provide insights into how to tackle the next step of the solution. Thus, by the end of this exploration, we should have solved four integrals and grasped a deeper understanding of the techniques required to approach this kind of problems.

First Integral

To start with our first integral, and first two terms, we'll attempt to solve the following:

$$\int_0^{\pi} \left(\frac{sin2x}{sinx}\right)^2 dx$$

At first glanced, the level of difficulty does not appear complex at all. We use the following double angle formula :

$$\sin 2x \ \equiv 2sinxcosx$$

$$\therefore \int_0^{\pi} \left(\frac{sin2x}{sinx}\right)^2 dx = \int_0^{\pi} \left(\frac{2sinxcosx}{sinx}\right)^2 dx = \int_0^{\pi} (2cosx)^2\, dx = \int_0^{\pi} 4\cos^2 x \ \ dx$$

Now, it is natural to consider applying the identity :

$$cos2x \equiv 2cos^2x - 1 \ \Leftrightarrow \ \cos^2 x \ \equiv \frac{cos2x + 1}{2}$$

$$\therefore \int_0^{\pi} 4\cos^2 x \ \ dx = \int_0^{\pi} 4\left(\frac{cos2x + 1}{2}\right) dx = \int_0^{\pi} (2cos2x + 2) \ \ dx$$

$$= \int_0^{\pi} 2cos2x\, dx + \int_0^{\pi} 2 \ \ dx \ = [sin2x + 2x]_0^{\pi}$$

$$= (\sin(2\pi) + 2\pi) - \big(\sin(0) + 2(0)\big) = 2\pi$$

Our first integral was pretty straightforward. Yet, it revealed key insights into the use of trigonometric identities, which we will likely need to be reconsidered throughout this exploration.

Second Integral

This time we'll try to solve our second integral, let it be :

$$I = \int_0^\pi \left(\frac{sin2x \cdot sin3x}{sinx \cdot sin6x}\right)^2 dx$$

Our first step is the same as our first integral, we simply by using the double angle formula of $sin2x$ and $sin6x$:

$$I = \int_0^\pi \left(\frac{sin2x \cdot sin3x}{sinx \cdot sin6x}\right)^2 dx = \int_0^\pi \left(\frac{2sinxcosx \cdot sin3x}{sinx \cdot 2sin3xcos3x}\right)^2 dx = \int_0^\pi \left(\frac{cosx}{cos3x}\right)^2 dx$$

This first step was pretty easy to see. But from now on, it is up to our intuition to find a way to solve it. So, the next thing I was tempted to do was to write $cos3x$ in terms of x, so that would allow us to remove one term from the equation.

Using the double angle formula, that gives us:

$$cos3x \equiv \cos(2x + x) \equiv cos2xcosx - sin2xsinx$$

and using the identities:

$$cos2x \equiv \cos^2 x - \sin^2 x$$

and

$$\cos^2 x \equiv 1 - \sin^2 x$$

thus, we have:

$$cos3x \equiv cosx(\cos^2 x - \sin^2 x) - 2\sin^2 x\, cosx$$

$$cos3x \equiv cosx(1 - 2\sin^2 x) - 2\sin^2 x\, cosx$$

$$cos3x \equiv cosx(1 - 4\sin^2 x)$$

We can substitute that into our integral:

$$I = \int_0^\pi \left(\frac{cosx}{cos3x}\right)^2 dx = \int_0^\pi \left(\frac{cosx}{cosx(1 - 4\sin^2 x)}\right)^2 dx = \int_0^\pi \frac{1}{(1 - 4\sin^2 x)^2}\, dx$$

From here, attempts to solve this new integral were numerous before finding an adequate solution, but among them, two stood out in particular. The one that will be carried out during this exploration consists of using the following substitution:

$$sin\,x \equiv \frac{tanx}{secx}$$

Indeed, we can verify that identity:

$$sinx \equiv \frac{tanx}{secx} \equiv \frac{\frac{sinx}{cosx}}{\frac{1}{cosx}} \equiv \frac{sinxcosx}{cosx} \equiv sinx$$

The other substitution we could have used is:

$$\sin x \equiv \frac{1}{\csc x}$$

Now, let's get back to our first substitution:

$$I = \int_0^\pi \frac{1}{(1-4\sin^2 x)^2}\,dx = \int_0^\pi \frac{1}{\left(1-4\frac{\tan^2 x}{\sec^2 x}\right)^2} dx$$

$$= \int_0^\pi \frac{1}{\left(\frac{\sec^2 x}{\sec^2 x} - \frac{4\tan^2 x}{\sec^2 x}\right)^2} dx = \int_0^\pi \frac{1}{\frac{(\sec^2 x - 4\tan^2 x)^2}{(\sec^2 x)^2}}\,dx$$

$$= \int_0^\pi \frac{(\sec^2 x)^2}{(\sec^2 x - 4\tan^2 x)^2} dx$$

The next step is using the identity:

$$\sec^2 x \equiv \tan^2 x + 1$$

That we can verify by starting from the Pythagorean theorem:

$$\cos^2 x + \sin^2 x \equiv 1$$

And then divide all terms by $\cos^2 x$, which gives us:

$$\frac{\cos^2 x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x} \equiv \frac{1}{\cos^2 x} \;\Leftrightarrow 1 + \tan^2 x \equiv \sec^2 x$$

At first glance, it's hard to understand how these substitutions can help us simplify our integral, but it's at our next step that we can start to notice it. If we substitute our last identity from our denominator and our numerator, we obtain:

$$I = \int_0^{\pi} \frac{(\sec^2 x)^2}{(\sec^2 x - 4\tan^2 x)^2} dx \;= \int_0^{\pi} \frac{\sec^2 x(\tan^2 x + 1)}{(1 - 3\tan^2 x)^2} dx$$

Now we clearly see it. Since $\sec^2 x$ is the derivative of $tan\, x$, a way to drastically simplify our integral requires the following substitution.

$$u = tanx \;\Rightarrow \frac{du}{dx} = \sec^2 x$$

Therefore, we obtain:

$$I = \int_b^a \frac{u^2 + 1}{(1 - 3u^2)^2} du, \qquad (a = tan(\pi)\,, b = tan(0) \;\Rightarrow\; \int_0^0 f(u)du = 0)$$

Again here, I think there are two different approaches on how to solve it. We could proceed by partial fraction directly since the degree of the denominator's polynomial is greater than the degree of the numerator's polynomial.

Or we can rewrite $u^2 + 1$ in terms of $(1 - 3u^2)$. So that it simplifies the integral. We'll still have to use partial fractions but those are going to be easier to calculate. In all cases, the task turns out to be long but doable. During my attempt to solve the integral, it is this second option that I used.

First step, as said earlier, we need to rewrite our numerator:

$$u^2 + 1 = -\frac{1}{3}(1 - 3u^2) + \frac{4}{3}$$

which gives us,

$$I = \int \frac{-\frac{1}{3}(1 - 3u^2) + \frac{4}{3}}{(1 - 3u^2)^2} du = \int \frac{-\frac{1}{3}(1 - 3u^2)}{(1 - 3u^2)^2} du + \int \frac{\frac{4}{3}}{(1 - 3u^2)^2} du$$

$$= \frac{4}{3}\int \frac{1}{(1 - 3u^2)^2} du - \frac{1}{3}\int \frac{1}{1 - 3u^2} du$$

Now that our integral is split in two others, we can solve them one by one.

$$let\ \ I_1 = \int \frac{1}{1-3u^2}du$$

When I first did it, I multiplied by 3 each term of the fraction. Looking back, it was an unnecessary thing to do, we could have directly applied partial fractions.

$$I_1 = \int \frac{3}{3-9u^2}du = \int \frac{3}{(\sqrt{3}+3u)(\sqrt{3}-3u)}du$$

Now we can apply partial fractions

$$I_1 = \int \frac{A}{\left(\sqrt{3}+3u\right)} + \frac{B}{(\sqrt{3}-3u)}du$$

Which allows us to obtain the following equation:

$$A\left(\sqrt{3}-3u\right) + B\left(\sqrt{3}+3u\right) = 3 \Leftrightarrow \sqrt{3}A - 3uA + \sqrt{3}B + 3uB = 3$$

From that equation, we can determine two others:

$$3uB - 3uA = 0 \Rightarrow A = B$$

and

$$\sqrt{3}A + \sqrt{3}B = 3\ \ \Leftrightarrow A + B = \frac{3}{\sqrt{3}} = \ \sqrt{3}$$

Since we just found with our previous equation that $A = B$, that gives us:

$$A = B = \frac{\sqrt{3}}{2}$$

Now going back to our integral, we obtain:

$$I_1 = \int \frac{\frac{\sqrt{3}}{2}}{(\sqrt{3}+3u)} + \frac{\frac{\sqrt{3}}{2}}{(\sqrt{3}-3u)}du = \frac{\sqrt{3}}{2}\int \frac{1}{\left(\sqrt{3}+3u\right)}du + \frac{\sqrt{3}}{2}\int \frac{1}{(\sqrt{3}-3u)}du$$

We can now solve these two integrals independently:

$$let\ H_1 = \frac{\sqrt{3}}{2}\int \frac{1}{\sqrt{3}+3u}du$$

This integral is pretty straightforward using a substitution:

$$let\ v = \sqrt{3} + 3u \ \Rightarrow \frac{dv}{du} = 3$$

$$H_1 = \frac{\sqrt{3}}{2} \cdot \frac{1}{3} \int \frac{1}{v} dv$$

$$H_1 = \frac{\sqrt{3}}{6} \ln|v| + C$$

And now, we need to solve the second one using the same principle:

$$let\ H_2 = \frac{\sqrt{3}}{2} \int \frac{1}{\sqrt{3} - 3u} du$$

$$w = \sqrt{3} + 3u \ \Rightarrow \frac{dw}{du} = -3$$

$$H_2 = -\frac{\sqrt{3}}{2} \cdot \frac{1}{3} \int \frac{1}{w} dw$$

$$H_2 = -\frac{\sqrt{3}}{6} \ln|w| + C$$

Combining them together:

$$I_1 = H_1 + H_2 = \frac{\sqrt{3}}{6} \ln|v| - \frac{\sqrt{3}}{6} \ln|w| + C = \frac{\sqrt{3}}{6} \left(\ln\left|\frac{v}{w}\right| \right) + C$$

Substituting back our variables v and w:

$$= \frac{\sqrt{3}}{6} \left(\ln\left|\frac{\sqrt{3} + 3u}{\sqrt{3} - 3u}\right| \right) + C$$

And finally, by substituting back our variable u, we get:

$$= \frac{\sqrt{3}}{6} \left(\ln\left|\frac{\sqrt{3} + 3tanx}{\sqrt{3} - 3tanx}\right| \right) + C$$

Until now, we have only solved the first part (I_1) of our initial integral (I). Before moving on to the next one, there are two ways to approach the second part of our integral: either we assume it will be more complicated because of the squared denominator, or we recognize that the pattern from the first part is similar since the denominator has a familiar form. And therefore, we should be able to solve it through the use of partial fractions.

$$let\ I_2 = \int \frac{1}{(1-3u^2)^2} du = \int \frac{1}{\left(1-\sqrt{3}u\right)^2\left(1+\sqrt{3}u\right)^2} du$$

$$= \int \frac{A}{\left(1-\sqrt{3}u\right)^2} + \frac{B}{1-\sqrt{3}u} + \frac{C}{\left(1+\sqrt{3}u\right)^2} + \frac{D}{1+\sqrt{3}u} du$$

$$A\left(1+\sqrt{3}u\right)^2 + B\left(1-\sqrt{3}u\right)\left(1+\sqrt{3}u\right)^2 + C\left(1-\sqrt{3}u\right)^2 + D\left(1+\sqrt{3}u\right)\left(1-\sqrt{3}u\right)^2 = 1$$

This time our equation doesn't simplify as easily as previously, so we need to find a new way in order to determine our variables $A, B, C\ and\ D$.

By using the following substitutions successively:

$$u = \frac{1}{\sqrt{3}}, u = -\frac{1}{\sqrt{3}}, u = \sqrt{3}\ \ and\ \ u = -\sqrt{3}$$

we find that:

$$A = B = C = D = \frac{1}{4}$$

Substituting that into our integral, we get:

$$I_2 = \frac{1}{4}\int \frac{1}{1-\sqrt{3}u} du + \frac{1}{4}\int \frac{1}{1+\sqrt{3}u} du + \frac{1}{4}\int \frac{1}{\left(1+\sqrt{3}u\right)^2} du + \frac{1}{4}\int \frac{1}{\left(1-\sqrt{3}u\right)^2} du$$

Then, using the following substitutions correctly as we did in the first part:

$$q = 1-\sqrt{3}u\ and\ r = 1+\sqrt{3}u$$

It gives us:

$$I_2 = \frac{1}{4\sqrt{3}}\int -\frac{1}{q^2} dq + \frac{1}{4\sqrt{3}}\int \frac{1}{r} dr - \frac{1}{4\sqrt{3}}\int \frac{1}{q} dq - \frac{1}{4\sqrt{3}}\int -\frac{1}{r^2} dr$$

$$= \frac{1}{4\sqrt{3}q} + \frac{\ln|r|}{4\sqrt{3}} - \frac{\ln|q|}{4\sqrt{3}} - \frac{1}{4\sqrt{3}r} + C$$

Substituting back all variables, we finally get:

$$I_2 = \frac{1}{4\sqrt{3}(1-\sqrt{3}tanx)} - \frac{1}{4\sqrt{3}(1+\sqrt{3}tanx)} + \frac{\ln\left|1+\sqrt{3}tanx\right|}{4\sqrt{3}} - \frac{\ln\left|1-\sqrt{3}tanx\right|}{4\sqrt{3}} + C$$

$$= \frac{tanx}{2(1-3\tan^2 x)} + \frac{\ln\left|\frac{1+\sqrt{3}tanx}{1-\sqrt{3}tanx}\right|}{4\sqrt{3}} + C$$

Lastly, our initial integral becomes:

$$I = \frac{4}{3}I_2 - \frac{1}{3}I_1 = \frac{4}{3}\left(\frac{tanx}{2(1-3\tan^2 x)} + \frac{\ln\left|\frac{1+\sqrt{3}tanx}{1-\sqrt{3}tanx}\right|}{4\sqrt{3}}\right) - \frac{1}{3}\cdot\frac{\sqrt{3}}{6}\left(\ln\left|\frac{\sqrt{3}+3tanx}{\sqrt{3}-3tanx}\right|\right) + C$$

To finish, we calculate the result of that bounded integral:

$$I = \int_0^{\pi}\left(\frac{sin2x\cdot sin3x}{sinx\cdot sin6x}\right)^2 dx$$

$$= \left[\frac{4}{3}\left(\frac{tanx}{2(1-3\tan^2 x)} + \frac{\ln\left|\frac{1+\sqrt{3}tanx}{1-\sqrt{3}tanx}\right|}{4\sqrt{3}}\right) - \frac{1}{3}\cdot\frac{\sqrt{3}}{6}\left(\ln\left|\frac{\sqrt{3}+3tanx}{\sqrt{3}-3tanx}\right|\right)\right]_0^{\pi}$$

We don't need to simplify further our equation since we can notice that $\tan\pi$ and $\tan 0$ are both equal to 0. In other words, all terms in our equation will be 0 and thus, will give a result of 0.

Now, we should verify our result by plotting the graph into GeoGebra.

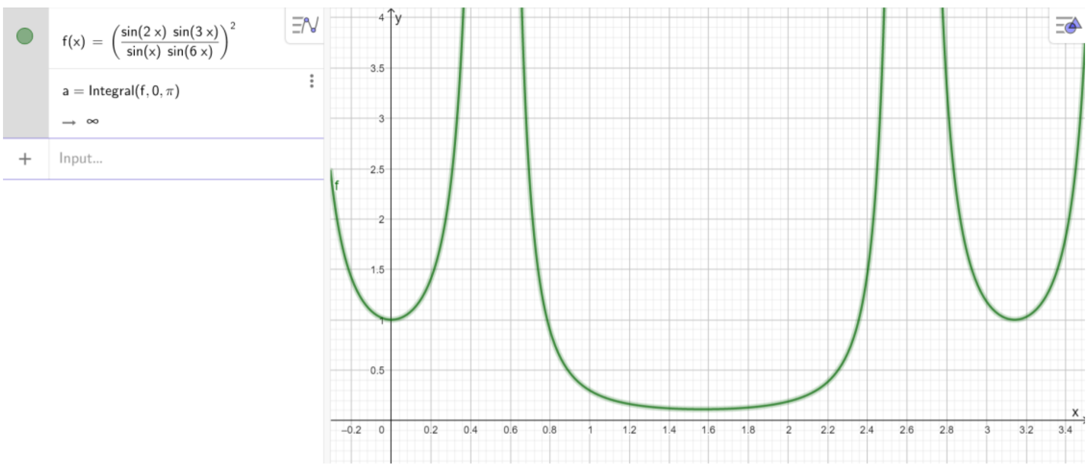

Figure 2 - Second Integral

However, the graph suggests the result is infinite. So, is our solution or the graph's solution that is wrong? First of all, we can see that the function has two asymptotes at $\frac{\pi}{6}$ and at $\frac{5\pi}{6}$ in the range $[0, \pi]$. That is why the area should effectively be infinite. So, is our solution wrong? To determine this, we can use our formula for the values at either side of these asymptotes, and it will give infinitely large numbers, confirming that the area is infinite. Also, by solving the integral for an interval from 0 to $\frac{\pi}{12}$ (before the asymptote) using the graph and using our formula, it gives the same result. Thus, our solution is also correct. Since our formula is right, why does it give 0 instead of ∞? Well, there are few possible explanations for that discrepancy. We can now exclude the possibility of negative and positive areas cancelling each other out since the function does not admit negative areas. The other reason that we can consider lies in the fact that when we're trying to solve integrals, and equations in general, we often make substitutions or use equivalence. However, when touching at deeper mathematical concepts, equivalent equations need to be used very carefully as it can remove a singularity that was in the original equation or ignore a point where the function is not defined. For instance:

The equation:

$$\frac{x^2 - 1}{x - 1} = x + 1, \quad x \neq 1$$

And the equation:

$$x^2 - 1 = (x + 1)(x - 1)$$

Seem algebraically equivalent. However, there is a hidden issue. Our initial equation was undetermined for $x = 1$, but our new equation just ignored this singularity. In other words, algebraic equivalence does not always imply analytical equivalence. Especially in our case, where trigonometric identities are likely to introduce and exacerbate this problem.

Finally, we can conclude that:

$$I = \infty$$

(The function must be continuous on a certain interval, to integrate that function on this interval, here we have an improper integral). Differentiability needs continuity and smoothness of the function.

This second integral seemed to be interminable. Having to resort to partial fractions might indicate to us that this technique will likely be used later on in our next steps of the exploration. We also discovered the necessity of applying integration by substitution, this technique could be key to our next findings. The pattern of trigonometric identities is once again present in the method we use to solve our integral. However, the level of difficulty begins to significantly increase as we add more terms to the integral, but it remains manageable. Furthermore, as we continue to add terms, the length of our approach may become even greater. Therefore, finding a more elegant solution, if it exists, will certainly be essential.

Third Integral

$$\int_0^{\pi} \left(\frac{sin2x \cdot sin3x \cdot sin5x}{sinx \cdot sin6x \cdot sin10x}\right)^2 dx$$

The first step remains the same. Simplifying we get:

$$= \int_0^{\pi} \left(\frac{cosx}{2cos3xcos5x}\right)^2 dx$$

Afterwards, any further step becomes significantly harder to make. Thereby, to conduct a decent investigation, it is key to re-assess the need to solve our third integral. At this point, we found some insights regarding the potential methods to use to solve our final integral. However, apart from simplifying with trigonometric identities, we cannot deduce any general pattern that would apply to all of our integrals. Following what has just been said and considering the myriad of unsuccessful attempts on my part to solve this integral, it would be more judicious to head directly towards our ultimate goal.

Fourth Integral

The ultimate aim of this exploration is to solve the following integral, given all the patterns we discovered along the way:

$$\int_0^{\pi} \left(\frac{sin2x \cdot sin3x \cdot sin5x \cdot sin30x}{sinx \cdot sin6x \cdot sin10x \cdot sin15x}\right)^2 dx$$

The first step remains the same (double angle formula for sin x):

$$= \int_0^{\pi} \left(\frac{2sinxcosx \cdot sin3x \cdot sin5x \cdot 2sin15xcos15x}{sinx \cdot 2sin3xcos3x \cdot 2sin5xcos5x \cdot sin15x}\right)^2 dx$$

Simplifying:

$$= \int_0^{\pi} \left(\frac{cosx \cdot cos15x}{cos3x \cdot cos5x}\right)^2 dx$$

From this integral, there are certainly many ways to simplify it further, I think it is up to your creativity and rational intuition to choose the way you do it. As I was guided by the numbers (3, 5 and 15), I first thought to use the double angle formula for $cos3x\ and\ cos15x$.

$$cos3x \equiv cos2xcosx - sin2xsinx \equiv cosx(\cos^2 x - \sin^2 x) - 2\sin^2 x\, cosx$$

$$\equiv cosx(1 - \sin^2 x - \sin^2 x - 2\sin^2 x) \equiv \cos x\,(1 - 4\sin^2 x)$$

$$cos15x \equiv cos10xcos5x - sin10xsin5x \equiv cos10xcos5x - 2\sin^2 5x\, cos5x$$

$$\equiv \cos 5x\,(\cos^2 5x - \sin^2 5x - 2\sin^2 5x) \equiv cos5x(1 - 4\sin^2 5x)$$

It is when I found out that these two expansions had a similar form, that I knew I was on the right path. So, let's plot our findings:

$$\int_0^{\pi} \left(\frac{cosx \cdot cos15x}{cos3x \cdot cos5x}\right)^2 dx = \int_0^{\pi} \left(\frac{cosx \cdot cos5x(1 - 4\sin^2 5x)}{cosx(1 - 4\sin^2 x) \cdot cos5x}\right)^2 dx$$

$$= \int_0^{\pi} \left(\frac{(1 - 4\sin^2 5x)}{(1 - 4\sin^2 x)}\right)^2 dx$$

Using the identity : $\sin^2 x \equiv 1 - \cos^2 x$, we get:

$$\int_0^\pi \left(\frac{4\cos^2 5x - 3}{4\cos^2 x - 3}\right)^2 dx$$

It is once I reach this point that I struggle the most to find the next steps to solve this integral. My teacher had a try at it and suggested using the double angle formula of $cos\ x$, and then resorting to complex numbers. Since it will involve expressions with exponents, which are easier to deal than trigonometric functions.

Firstly, we use the following identity:

$$\cos^2\theta \equiv \frac{(cos2\theta + 1)}{2}$$

We can also apply it for our numerator:

$$by\ letting\ \theta = 5x \Rightarrow \cos^2 5x \equiv \frac{(\cos 10x + 1)}{2}$$

$$\therefore \int_0^\pi \left(\frac{4\cos^2 5x - 3}{4\cos^2 x - 3}\right)^2 dx = \int_0^\pi \left(\frac{4\left(\frac{cos10x + 1}{2}\right) - 3}{4\left(\frac{cos2x + 1}{2}\right) - 3}\right)^2 dx = \int_0^\pi \left(\frac{2cos10x - 1}{2cos2x - 1}\right)^2 dx$$

Now, this form might appear familiar if we are accustomed to solving problems of this kind. The next step would be to resort to Euler's formula:

$$e^{ikx} = \cos(kx) + isin(kx)$$

$$e^{-ikx} = \cos(-kx) + isin(-kx) = \cos(kx) - isin(kx)$$

$$\Rightarrow e^{ikx} + e^{-ikx} = 2\cos(kx)$$

$$\therefore \int_0^\pi \left(\frac{2cos10x - 1}{2cos2x - 1}\right)^2 dx = \int_0^\pi \left(\frac{e^{10ix} + e^{-10ix} - 1}{e^{2ix} + e^{-2ix} - 1}\right)^2 dx$$

Seeing our new equation, one might wonder why we went through all that. In fact, our substitution made it look more complex to integrate than it was before. But sometimes in mathematics, you need to introduce a more complicated form to uncover the underlying simplicity. Indeed, by going further, we can use the following substitution which will ultimately help in solving our integral:

$$let\ p = e^{ix}$$

Thus, our integral becomes:

$$\int_0^{\pi} \left(\frac{p^{10}+p^{-10}-1}{p^2+p^{-2}-1}\right)^2 dx$$

We can distinguish the form of one polynomial dividing the other. However, a synthetic division cannot be used for polynomials with negative exponents. If we attempted it, we would end up with a polynomial having infinitely many terms, each smaller than the previous one. Therefore, there are two possible ways of approaching it. The first one involves the division of the polynomials with negative exponents, also known as Laurent polynomials, they have their own division rules and properties, a technique for dividing them is called the *long division,* which is not more difficult than synthetic division. Alternatively, we could rewrite our polynomials in terms of positive exponents as follow, and then apply synthetic division. This second option is preferable since synthetic division is within the scope of the IB Math AA curriculum.

$$\int_0^{\pi} \left(\frac{p^{-10}(p^{20}-p^{10}+1)}{p^{-2}(p^4-p^2+1)}\right)^2 dx$$

$$\int_0^{\pi} \left(\frac{1}{p^8}\cdot\frac{(p^{20}-p^{10}+1)}{(p^4-p^2+1)}\right)^2 dx$$

Now, the synthetic division process is detailed in the Appendix 1 and gives the following result:

$$\frac{(p^{20}-p^{10}+1)}{(p^4-p^2+1)} \equiv p^{16}+p^{14}-p^{10}-p^8-p^6+p^2+1$$

Plotting that into our integral:

$$\int_0^{\pi} \left(\frac{p^{16}+p^{14}-p^{10}-p^8-p^6+p^2+1}{p^8}\right)^2 dx$$

$$= \int_0^{\pi} (p^8+p^6-p^2-1-p^{-2}+p^{-6}+p^{-8})^2\, dx$$

Here, I expanded the brackets on paper as shown in Appendix 2 because I wanted to stay true to the format of the competition, which forbids the use and access to any technology.

However, we could have simply used a framework to do it for us, as well as performing the synthetic division. So, after the expansion, we obtain:

$$\int_0^{\pi} (p^{16} + p^{-16} + 2p^{14} + 2p^{-14} + p^{12} + p^{-12} - 2p^{10} - 2p^{-10} - 4p^8 - 4p^{-8} - 4p^6 - 4p^{-6} - p^4 - p^{-4} + 4p^2 + 4p^{-2} + 7)dx$$

Substituting back our variable p, and directly simplifying our integral, we finally get:

$$= \int_0^{\pi} (2cos16x + 4cos14x + 2cos12x - 4cos10x - 8cos8x - 8cos6x - 2cos4x + 8cos2x + 7)dx$$

$$= \left[\frac{sin16x}{8} + \frac{4sin14x}{14} + \frac{sin12x}{6} - \frac{4sin10x}{10} - sin8x - \frac{8sin6x}{6} - \frac{sin4x}{2} + 4sin2x + 7x\right]_0^{\pi}$$

Here, like our second integral, all our sine terms become zero, since $sin(\pi) = sin(0) = 0$. Thus, we are left with the following:

$$[7(\pi) - 7(0)] = 7\pi$$

To verify our findings, the framework GeoGebra can estimate approximatively the result of our integral:

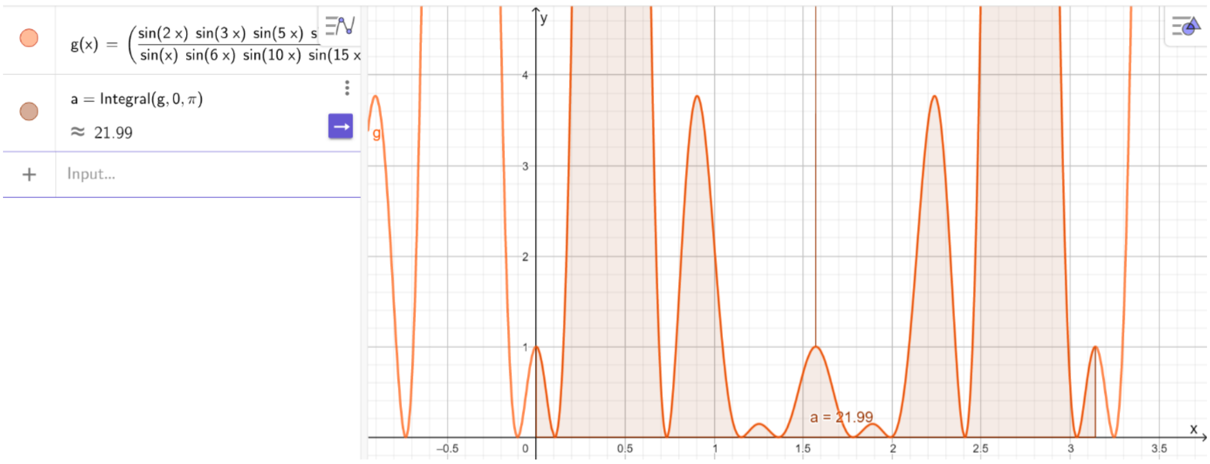

Figure 3 - Graph of the fourth integral

For instance, it evaluated our integral to be $\approx 21.99 \approx 7\pi$. This corroborates the accuracy of our result.

Conclusion

The aim of this exploration was to solve our fourth integral, which we successfully achieved. Initially, we aimed to identify patterns or gain insights from the first three integrals. We observed that trigonometric identities were omnipresent, whereas partial fractions were only necessary for the second integral. This second integral required a lot of substitutions, whereas the fourth one involved significantly fewer. Although we attempted to find repetitive solving patterns across all four integrals, we ultimately concluded that the fourth integral required completely different techniques. In the end, we acknowledge that solving the fourth integral was a lengthy process, making it surprising that candidates were given only four minutes to complete it. However, given the high level of requirements at MIT, it is perhaps not that surprising that this problem was expected to be solved within such a short amount of time. It is also possible that an even more elegant and simpler solution was anticipated.

Appendix 1 – Synthetic Division

$$
\begin{array}{r|l}
 & p^{16}+p^{14}-p^{10}-p^{8}-p^{6}+p^{2}+1 \\
\hline
p^{4}-p^{2}+1 & p^{20}-p^{10}+1 \\
 & -(p^{20}-p^{18}+p^{16}) \\
\cline{2-2}
 & p^{18}-p^{16}-p^{10}+1 \\
 & -(p^{18}-p^{16}+p^{14}) \\
\cline{2-2}
 & -p^{14}-p^{10}+1 \\
 & -(-p^{14}+p^{12}-p^{10}) \\
\cline{2-2}
 & -p^{12}+1 \\
 & -(-p^{12}+p^{10}-p^{8}) \\
\cline{2-2}
 & -p^{10}+p^{8}+1 \\
 & -(-p^{10}+p^{8}-p^{6}) \\
\cline{2-2}
 & p^{6}+1 \\
 & -(p^{6}-p^{4}+p^{2}) \\
\cline{2-2}
 & p^{4}-p^{2}+1 \\
 & -(p^{4}-p^{2}+1) \\
\cline{2-2}
 & 0
\end{array}
$$

Appendix 2 – Brackets expansion

$$(\rho^{8}+\rho^{6}-\rho^{2}-1-\rho^{-2}+\rho^{-6}+\rho^{-8})^{2} =$$

$$(\rho^{8}+\rho^{6}-\rho^{2}-1-\rho^{-2}+\rho^{-6}+\rho^{-8})(\rho^{8}+\rho^{6}-\rho^{2}-1-\rho^{-2}+\rho^{-6}+\rho^{-8}) =$$

$$\begin{aligned}
&\rho^{16}+\rho^{14}-\rho^{10}-\rho^{8}-\rho^{6}+\rho^{2}+1 && (1)\\
&+\rho^{14}+\rho^{12}-\rho^{8}-\rho^{6}-\rho^{4}+1+\rho^{-2} && (2)\\
&-\rho^{10}-\rho^{8}+\rho^{4}+\rho^{2}+1-\rho^{-4}-\rho^{-6} && (3)\\
&-\rho^{8}-\rho^{6}+\rho^{2}+1+\rho^{-2}-\rho^{-6}-\rho^{-8} && (4)\\
&-\rho^{6}-\rho^{4}+1+\rho^{-2}+\rho^{-4}-\rho^{-8}-\rho^{-10} && (5)\\
&+\rho^{2}+1-\rho^{-4}-\rho^{-6}-\rho^{-8}+\rho^{-12}+\rho^{-14} && (6)\\
&+1+\rho^{-2}-\rho^{-6}-\rho^{-8}-\rho^{-10}+\rho^{-14}+\rho^{-16} && (7)
\end{aligned}$$

$$\boxed{\begin{aligned}
&=\rho^{16}+\rho^{-16}+2\rho^{14}+2\rho^{-14}+\rho^{12}+\rho^{-12}-2\rho^{10}-2\rho^{-10}-4\rho^{8}-4\rho^{-8}-4\rho^{6}-4\rho^{-6}\\
&-\rho^{4}-\rho^{-4}+4\rho^{2}+4\rho^{-2}+7
\end{aligned}}$$